The Department of Geological Engineering is part of the Faculty of Engineering and Architecture and is officially accredited by the Pakistan Engineering Council. It was initiated in 2008 as the first degree-awarding Geological Engineering program under the engineering faculty at BUITEMS. It was established to address the critical need for hidden natural resource exploration and economic development in Balochistan, which is Pakistan's largest and most mineral and hydrocarbon-rich province, with roughly ninety percent of its area historically unexplored. The core technical domains involve analyzing the engineering behavior of rock, soil, and geotechnics against constructed projects, and guiding mineral and hydrocarbon discovery. Graduates are in high demand in the energy and construction sectors, finding roles as geotechnical engineers, geologists, construction consultants, or civil contractors across public and private organizations. Key employment fields include petroleum and gas organizations, atomic energy, the Irrigation Department, academic institutions, and Natural Energy Resources Exploration. The mission of the department is committed to train engineers with advanced knowledge and contemporary skills, while promoting a culture of research and equipping students with the art of living to achieve success in their future professional careers not only in Pakistan but throughout the world. Graduates are expected to demonstrate specific program educational objectives three to five years post-graduation. These include the ability to apply technological and engineering fundamentals for the solution of geotechnical and environmental problems related to geological processes, the ability to exhibit effective communication, teamwork, leadership skills, and technical competence enabling them to excel in their professional careers and higher studies, and the ability to fulfill ethical, professional, and social responsibilities in their professional lives and community services.

For the Bachelor of Science in Geological Engineering, the admission requirements mandate an academic background of F.Sc Pre-Engineering, Pre-Medical, or ICS from any recognized board with at least sixty percent marks, or a Diploma of Associate Engineer in Civil, Land and Mine Surveying, or Mining securing a minimum of sixty percent marks. Candidates must also successfully qualify for the university admission test. The degree requirements encompass a total of 136 credit hours across 46 courses, and students must maintain a minimum academic standing with a CGPA of 2.0 or higher. The program adopts the twelve expected standard engineering graduate attributes defined in the Pakistan Engineering Council Accreditation Manual 2014, which include Engineering Knowledge, Problem Analysis, Design and Solutions, Investigation, Modern Tool Usage, Engineer and Society, Sustainability, Ethics, Teamwork, Communication, Project Management, and Lifelong Learning.

The curriculum plan outlines the program schema by semester, where credit hour structures denote theory plus lab hours. In the first semester, students take Engineering Drawing and Graphics or Basic Electrical and Electronics for one plus one or two plus one credit hours, Physical Geology or Applied Chemistry for three plus one or two plus zero credit hours, Calculus and Analytical Geometry or Islamic Studies or Ethics for three plus zero or two plus zero credit hours, and Functional English for two plus zero credit hours. The second semester includes Mineralogy and Petrology or Introduction to Geological Engineering for two plus one or two plus zero credit hours, Differential Equations or Applied Physics for three plus zero or two plus zero credit hours, Communication Skills or Pak-Studies for two plus zero credit hours each, and Introduction to ICT or Occupational Health and Safety for two plus lab or one plus zero credit hours, where the lab component for ICT is noted as incomplete in the raw text list. The third semester requires Applied Thermodynamics or Stratigraphy and Structural Geology for two plus one credit hours each, Linear Algebra or Fluid Mechanics for three plus zero or three plus one credit hours, and Computer Programming or Engineering Economics for two plus one or two plus zero credit hours. In the fourth semester, students take Mechanics of Material or Surveying for three plus one or two plus two credit hours, Computer Science and Numerical Analysis or Hydrogeology for two plus zero or three plus one credit hours, and Probability and Statistics or Technical Report Writing and Presentation Skills for two plus zero credit hours each.

The fifth semester consists of Engineering Geology or Introduction to Geophysical Exploration Tech for two plus one or three plus one credit hours, Geotechnical Engineering-I or Explosives Engineering for three plus one or two plus one credit hours, and Engineering Management or Sociology for Engineers for two plus zero credit hours each. The sixth semester covers Petroleum Geology or Rock Mechanics for two plus one or three plus one credit hours, Earth Quake Seismology and Risk Assessment or GIS and Remote Sensing for two plus one credit hours each, and Environmental Geological Engineering for two plus one credit hours. The seventh semester involves Major Elective I or Tunneling and Excavation Engineering for three plus one credit hours each, Major Elective II or Drilling Engineering for two plus one credit hours each, and Project-I for zero plus two credit hours. The eighth semester concludes the core requirements with Major Elective III or Major Elective IV for three plus one or two plus one credit hours, and Entrepreneurship or Project-II for three plus zero or zero plus four credit hours, bringing the grand total to 136 credit hours.

Under the fifth Board of Studies framework, the major electives curriculum offers twenty-four elective courses partitioned into three distinct professional specializations, which are Geotechnical and Rock Engineering, Natural Energy Resource Exploration, and Geo-Environmental Engineering. Students must choose four courses in total from one preferred stream, taking two in the seventh semester and two in the eighth semester. For Major Elective-I in the seventh semester for three plus one credits, options include Geotechnical Engineering-II or Statistical Methods in Geology and Engineering for the first stream, Seismic Data Processing and Interpretation or Ground water Resource Engineering for the second, and Environment Geology and Waste Management or Environmental Aspects of Mining for the third. For Major Elective-II in the seventh semester for two plus one credits, students can choose Geotechnical Site Investigation or Fluvial Processes, Reservoir Engineering or Surficial Processes, Sedimentation and Stratigraphy, or Risk Assessment in Environmental Studies or Exploration and Environmental Geochemistry. For Major Elective-III in the eighth semester for three plus one credits, the choices are Pavement and Foundation Engineering or Geomorphology and Terrain Analysis, Petrophysics and Well Logging or Petrogenesis and Metallogenesis, and Environmental Soil Science or Site Remediation Engineering. Finally, for Major Elective-IV in the eighth semester for two plus one credits, the options are Ground Improvement and Geosynthetics or Analysis of Rock Structures, Production Engineering or Carbonate Sedimentology, and Environmental Data Analysis or Recycling and Sustainable Engineering.